

LatentEdge: The Behavioral Pattern Spec That Actually Ships

Bottom line up front. Retail traders fail at 70–93% rates across every market we studied not because they lack market knowledge but because they execute the same eight to ten repeating behavioral loops — and almost all of those loops are detectable from structured trade-log data alone. Of 22 distinct failure mechanisms identified in the literature, **10 should anchor V1, 4 should be merged into umbrella patterns, 5 should be deferred to V2** until cold-start data thickens, and **3 should be cut entirely as low-signal**. The minimum viable V1 detects 10 patterns from existing schema fields plus four small additions (planned stop loss, planned target, planned trigger price, account equity snapshot), uses **soft nudges by default and hard blocks for two specific triggers** (daily-loss-limit revenge entries and stop-loss removal), and is marketable as *"we surface the eight behaviors costing you 80% of your losses."*

The rest of this document delivers Steps 1–9 plus an executive recommendation layer that maps each pattern to schema fields, SQL-style detection, prop-firm rule consequences, and rollout sequence.

Step 1 — Core failure mechanisms as repeatable behavioral loops

Each loop below is the form **trigger → cognitive/affective response → action → outcome → reinforcement**, with the academic anchor and a reactive/proactive tag. These are *root* mechanisms, not symptoms.

#	Loop name	Trigger → Response → Action → Outcome → Reinforcement	Psychological driver	Type
1	Revenge-Loss Doubling	2 consecutive losses → frustration / "market owes me" → next entry at 2-3× size on marginal setup → larger loss → "cursed day" narrative; either capitulates or doubles again (Highkeyfx)	Break-even effect (Thaler & Johnson 1990); (Stanford) (Semantic Scholar) loss aversion (Kahneman-Tversky 1979)	Reactive
2	Stop-Widening Death Spiral	Price approaches stop → anticipatory pain of realizing loss → moves stop further → bigger realized loss → concludes "stops don't work for me" → next trade has no hard stop (TradeZella)	Loss aversion; illusion of control (Langer 1975)	Reactive (intra-trade)
3	Premature Profit-Taking (Disposition)	Trade reaches +0.3R to +0.5R → fear of giving back → exits early → trade hits planned target without him → regret → over-holds next loser to "prove patience"	Disposition effect (Odean 1998); (Haas School of Business) (Wiley Online Library) fear of leaving money on table (Douglas)	Proactive (structural)
4	Hot-Streak Oversizing	3 consecutive winners → euphoria, "house money" → 1.5-2× size on next setup without quality check → larger loss erases gains → freeze on next valid signal	House-money effect (Thaler-Johnson); (SSRN) (Semantic Scholar) overconfidence (Gervais-Odean 2001)	Reactive
5	FOMO Chase	Watches vertical move 10-20 min without entering → rising regret → enters at top tick on no signal → adverse excursion → cuts → trend resumes without him	Affect heuristic (Slovic et al.); attention-driven trading (Barber-Odean 2008)	Proactive
6	Break-Even Anchor Trap	Position underwater → "I'll wait until break-even" → holds through invalidation → small loss becomes catastrophic	Anchoring (Tversky-Kahneman 1974); regret aversion	Reactive (intra-trade)

#	Loop name	Trigger → Response → Action → Outcome → Reinforcement	Psychological driver	Type
7	Sunk-Cost Averaging-Down	Loser at -1R → "already lost this much, may as well add" → averages down → 2× exposure deteriorates → forced exit at -4R or liquidation (Aron Groups)	Sunk-cost fallacy; prospect-theory risk-seeking in losses	Reactive (intra-trade)
8	Hesitation-Then-Chase	Plan triggers at level X → freezes → price extends 1R without him → enters late → planned stop now mispriced → asymmetric loss	Fear of being wrong (Douglas); Steenbarger trigger-slippage	Proactive
9	Boredom Over-Trading	Quiet midday tape, no A-setups → restlessness → forces B/C-grade setup → loses → drops into revenge mode	Affect heuristic; illusion of control; dopamine-seeking	Proactive (state)
10	End-of-Day / End-of-Cycle Make-Back	30 min before close (or last day of prop-firm cycle) on red → urgency to "not finish red" → oversized low-prob trade → typically loses	Break-even effect; mental accounting (Thaler 1985)	Reactive
11	Disposition Inversion	Persistent: holds losers (avg -2.3R, 4.5h) and cuts winners (+0.6R, 0.7h) → expectancy collapses despite >50% hit rate → switches systems, same behavior repeats	Disposition effect (Odean 1998; Shefrin-Statman 1985) (Haas School of Business +2)	Proactive (structural)
12	Regime-Blind Persistence	Volatility regime shifts (trend → chop) → continues breakout playbook → 5+ losses in a row → doubles down on prior narrative instead of testing regime change	Narrative fallacy (Taleb); failure to adapt (Lo AMH) (Wikipedia)	Proactive
13	Recency-Driven System Abandonment	4 losses in 6 trades (within 50% system's expected variance) → "edge broken" → switches systems mid-drawdown → new system	Recency bias; small-sample fallacy (Tversky-Kahneman 1971)	Reactive (long horizon)

#	Loop name	Trigger → Response → Action → Outcome → Reinforcement	Psychological driver	Type
		mean-reverts and would have made next 5 winners → pattern repeats quarterly		
14	Leverage- Funding Trap (crypto/forex)	Funding rate or leverage in top decile → enters at peak crowding → liquidation cascade catches stop	Overconfidence; herding	Proactive
15	Theta- Gambling on Expiry	Long premium 0-2 DTE → option decays into close → exits at near-total loss; or short naked premium into scheduled event → IV gap- blowup	Lottery-preference / probability weighting (Kahneman- Tversky); house-money	Proactive

Loops 14 and 15 are market-specific expressions of the cross-market loops above; they earn their own line because their **detection logic and severity differ materially** in crypto and Indian F&O.

Step 2 — Mapping each failure to detectable signals in the schema

Detection columns reference the exact schema fields you defined. **Bold** means the field is in the existing schema; *italic* means an additive schema field (covered in Step 8).

Loop	Trade-level signal	Session-level signal	Sequence pattern	Psych metadata
1 Revenge	$\frac{\text{quantity} * \text{entry_price}}{\text{account_equity}}$ rises $\geq 1.5 \times$ rolling-30-trade median	≥ 2 trades within 30 min of prior loss in same <code>session</code>	Triggered after 2+ consecutive <code>win_loss=L</code>	<code>emotion_entry IN ('revenge', 'anxious')</code> <code>trade_type='impulsive'</code> <code>followed_plan='no plan'</code>
2 Stop-widen	<code>stop_loss_moved='widened'</code> ; <i>exit_price worse than initial planned stop</i>	Concentrates on red sessions	Often after small adverse move	<code>emotion_exit='hesitant'</code>
3 Premature exit	<code>r_multiple</code> realized $< \text{planned}_r$ by $\geq 40\%$; <code>hold_minutes</code> on win $< \text{median}(\text{hold_minutes} \text{loss}) \times 0.5$	Any	Wins consistently shorter than losers	<code>what_went_wrong</code> tag
4 Hot-streak oversize	<code>quantity</code> $\geq 1.3 \times$ rolling median after <code>win_loss=W</code> $\times 3$	Often last trade of streak day	After ≥ 3 consecutive wins	<code>emotion_entry='overconfident'</code> <code>conviction_level=3</code>
5 FOMO chase	<i>entry_price percentile in last 60-min bar range > 0.85 (long) / < 0.15 (short)</i> ; <code>setup_type</code> empty or "news_play"	Within session of a vertical move	Often no recent loss	<code>emotion_entry='fomo'</code> <code>trade_type='impulsive'</code> <code>followed_plan='no plan'</code>
6 Break-even anchor	<code>exit_price</code> within 0.1% of <code>entry_price</code> after MFE moved against; <code>hold_minutes</code> on losers $\gg$ winners	Any	Repeated exit clustering at entry price	<code>what_went_wrong</code> tag "breakeven"
7 Averaging-down	Multiple <code>trade_id</code> s same <code>instrument_name</code> + <code>direction</code> opened while prior leg has negative unrealized P&L (computed from sequence)	Same session usually	Within-position adds	<code>conviction_level</code> deteriorates
8 Hesitation-chase	$\text{abs}(\text{entry_price} - \text{planned_trigger_price}) > 0.5 \times \text{ATR}$; <code>trade_type='impulsive'</code>	Around session opens	Often follows skipped earlier signal	<code>emotion_entry='hesitant'</code> <code>'fomo'</code>
9 Boredom	<code>hold_minutes</code> short, <code>quantity</code> below median, <code>session='midday'</code> , <code>market_condition='ranging'</code>	Lunch-hour cluster (12:00-13:30 local)	Not after a loss	<code>emotion_entry='boredom'</code> <code>conviction_level=1</code>

Loop	Trade-level signal	Session-level signal	Sequence pattern	Psych metadata
	<code>setup_type</code> weak; trade frequency inverse to <i>vol_index</i>			
10 EOD make-back	<code>entry_time</code> in last 60 min of <code>session='closing'</code> ; <code>quantity</code> $\geq 1.5 \times$ median; cumulative day P&L < 0; or last 3 trading days of <i>prop_cycle</i>	Closing session, red days	After cumulative red day	<code>emotion_entry</code> IN ('anxious', 'revenge')
11 Disposition inversion	Account-level: $\text{avg}(\text{hold_minutes} \text{win}) / \text{avg}(\text{hold_minutes} \text{loss}) < 0.7$; PGR/PLR > 1.3	Persistent	All days	None needed
12 Regime-blind	Rolling-30-trade SQN dropping $\geq 30\%$ while <code>strategy_id</code> and <code>setup_type</code> unchanged; loss rate concentrated in one <code>market_condition</code>	Persistent	Long horizon	<code>followed_plan</code> ='yes'
13 System abandon	<code>strategy_id</code> change within 5 trades of >5% drawdown; <30 trades per strategy before switch	After local drawdown	Quarterly cluster	<code>notes</code> mentions "new strategy"
14 Leverage-funding	<code>leverage</code> in top decile of user history; <i>funding_rate</i> or <i>open_interest_pctile</i> from market context in top decile; <code>asset_class</code> IN ('crypto', 'forex')	News/macro events	Often after FOMO	<code>conviction_level</code> > 0.5
15 Theta-gambling	<code>asset_class</code> ='fno' or US options, <code>instrument_type</code> ='option', days-to-expiry ≤ 2 ; long premium in last 90 min OR naked short premium with notional >5 $\times$ collected, held into scheduled event	Thursday (NSE Nifty), Friday/expiry	Cyclical weekly	<code>setup_type</code> ='news'

Step 3 — Quantified business impact

Loop	Expectancy hit	R:R impact	Drawdown / blow-up risk	Prop-firm rule consequence
1 Revenge	Single trade can erase 3-5 days of edge; converts +0.2R/trade system into negative	Caps R:R at -2 to -5R	#1 cause of FTMO daily-loss-limit (5%) breaches per FTMO public commentary	Daily Loss Limit → instant fail
2 Stop-widen	Realized R 2-4× planned; turns -1R distribution into long left tail	Catastrophic	Single position can hit max-DD (10%)	Max Drawdown → instant fail
3 Premature exit	Disposition effect drag = 2-5% annualized (Odean 1998); (TradesViz +2) compounds over years	Compresses winners to <1R while losers run -2R	Slow bleed; rarely triggers prop limits but fails Phase-1 profit target	Misses 10% target → time/fee loss
4 Hot-streak oversize	Reverses a profitable streak; one trade ≈ 30-50% of monthly P&L volatility	Asymmetric -3R loss on size that won't be recovered at planned 1R	Trailing-DD breach post-funding	Trailing Max DD → funded acct lost
5 FOMO chase	Negative expectancy: entries in worst 10% of bar range underperform across studies	R:R inverted (entry near top)	Death-by-1000-cuts	Eats into daily buffer
6 Anchor	-1R becomes -3R+; converts swing system into binary	Loss tail >> planned	Major drawdown contributor	Max DD breach
7 Averaging-down	Position size 2× → losses 2×; canonical pre-blow-up signature	Catastrophic	Primary blow-up mechanism in grid/martingale; SEBI gap-day short-strangle wipeouts	Instant max-DD breach
8 Hesitation-chase	Fills 1R worse than planned; 30-50% expectancy haircut	R:R squeezed	Modest drawdown	Misses targets, prolongs evaluation

Loop	Expectancy hit	R:R impact	Drawdown / blow-up risk	Prop-firm rule consequence
9 Boredom	Negative-EV B/C trades; transaction-cost drag (SEBI: 27% of gross losses are costs for loss-makers) (Business Today) (Wright Research)	<1:1 R:R typical	Cost-driven attrition	Erodes daily buffer
10 EOD make-back	Single trade; converts breakeven day into max-loss day	-3 to -5R	Frequent on red Fridays / final prop-cycle days	DLL & cycle deadline failures
11 Disposition inversion	Direct 2-5% annual drag; can flip winning system to losing	Inverted (losers > winners)	Slow attrition	Fails profit target
12 Regime-blind	Persistent -ve expectancy until regime shifts back	Degraded	Drawdown 10-20% on previously winning system	Phase failure
13 System abandon	Locks in drawdowns; never realizes mean reversion of edge	N/A	Career-long underperformance	Continual evaluation churn
14 Leverage-funding	At 125× simulated, 98.3% liquidation rate (NYU Stern); (NYU Stern) Oct 2025 cascade liquidated (SSRN) ~\$20B retail (Solidus Labs)	Binary	Account wipeout in <30 min during cascades	N/A (funded firms cap leverage)
15 Theta-gambling	Long ODTE: - \$8.05/contract average (Beckmeyer 2023); (Medium) short premium gap risk on NSE Bank Nifty / event days catastrophic	Long: lottery; short: hidden tail	Naked short premium = single-day blow-up	Multi-day evaluation kill

Step 4 — Cross-market validation

Loop	Forex	Crypto	India F&O	NASDAQ/US
1 Revenge	✓ — high-leverage amplifies single-trade DLL hit	✓ — 24/7 means no cooling-off; Aug 2024 long-liquidation cluster	✓ — common on red Bank Nifty days; weekly expiry pressure	✓ — 0DTE accelerates revenge cycle to minutes
2 Stop-widen	✓ — esp. EAs that move SL programmatically	✓ — manual stop drag during cascades	✓ — Bank Nifty short premium "give it room"	✓
3 Premature exit	✓ — Odean baseline replicates	✓ — exacerbated by 24/7 vigilance	✓ — STT cost magnifies cumulative drag	✓
4 Hot-streak	✓	✓ — funding-rate top-decile entries	✓	✓
5 FOMO	Moderate — slower price action	High — memecoin/altcoin cycles, ~70% of altcoin retail entries within 10% of local highs	✓ — event-driven option buying (Budget, RBI)	High — meme stocks, AI narrative names
6 Anchor	✓	✓ — crypto entries near ATHs persist for years	✓	✓ — earnings underwater holds
7 Averaging-down	High — grid/martingale EAs canonical (Acy)	✓ — alts and leverage longs	High — gap-day short-premium	✓
8 Hesitation-chase	✓ — London/NY open	✓	✓ — opening 09:15-09:45 IST	✓ — opening 30 min
9 Boredom	✓ — Asian session chop	Severe — 24/7 always available	Moderate — bounded session 09:15-15:30	✓ — lunch hour 12:00-13:30 ET
10 EOD make-back	✓ — Friday close	✓ — every UTC midnight; or end of weekend	Severe — Thursday expiry close on Bank Nifty before reform; now Nifty Thursday	✓ — last-hour 0DTE

Loop	Forex	Crypto	India F&O	NASDAQ/US
11 Disposition inv.	✓ universal	✓	✓	✓
12 Regime- blind	✓ — esp. trend- following EAs in chop	✓ — 2022 mean- reversion vs 2023 trend	✓ — short-vol regime shift	✓
13 System abandon	✓ — "strategy hopping" documented FCA/ASIC	✓	✓	✓
14 Leverage- funding	✓ — offshore 500- 2000:1 vs ESMA 30:1 Investment Law Group	Critical — 125× perpetuals, arXiv \$20B Oct-2025 cascade Solidus Labs Substack	Limited — peak-margin rules cap; manifests as option-buying leverage proxy	Limited — Reg-T 4:1, but ODTE = synthetic leverage
15 Theta- gambling	N/A directly	Limited (deribit options small retail share)	Critical — Thursday Nifty expiry (post-Nov 2024 reform), single weekly expiry per exchange, Business Standard Wright Research lot size ₹15-20L Business Standard	Critical — ODTE >75% of retail SPX option flow SSRN +2

Differentiation insights for product copy: crypto FOMO + leverage = liquidation; Indian F&O = transaction-cost + theta gambling; forex = leverage abuse + EA grid blow-ups; NASDAQ = ODTE + meme attention.

Step 5 — Ranking by product relevance

Each pattern scored 1-5 on Detectability (D), Financial Impact (F), Emotional Recognition (E), Actionability (A). **Total = D + F + E + A** (max 20).

#	Loop	D	F	E	A	Total	Tier
1	Revenge	5	5	5	5	20	T1
2	Stop-widening	4*	5	4	5	18	T1
3	Premature exit (Disposition)	4	4	5	4	17	T1
7	Averaging-down	5	5	4	5	19	T1
4	Hot-streak oversize	5	4	4	5	18	T1
10	EOD make-back / cycle-end	5	4	4	5	18	T1
5	FOMO chase	4	4	5	4	17	T1
9	Boredom over-trading	4	3	4	4	15	T1
11	Disposition inversion (account)	5	4	4	3	16	T1
15	Theta gambling	5	5	3	4	17	T1 (market-gated)
14	Leverage-funding trap	4	5	3	3	15	T2
8	Hesitation-chase	3*	3	4	4	14	T2
6	Break-even anchor	4	4	4	3	15	T2 (merged)
12	Regime-blind persistence	3	4	2	2	11	T2
13	System abandonment	3	3	3	3	12	T2
16	Identity-defense hold	1	3	3	2	9	T3 (cut)
17	Trauma-anniversary freeze	1	2	4	1	8	T3 (cut)
18	Confirmation scroll	1	2	3	2	8	T3 (cut)
19	Hindsight-distorted review	2	2	3	2	9	T3 (cut)
20	Gambler's-fallacy mean-reversion	2	3	2	2	9	T3 (cut)
21	Hidden correlation across positions	4	3	2	3	12	T2
22	News-event slippage	4	4	2	4	14	T2 (merged into 14)

*Detectability rises to 5 with planned-stop and planned-trigger schema additions.

Result: 10 Tier-1, 6 Tier-2, 5 Tier-3.

Step 6 — Tier-1 patterns as product features

Each Tier-1 pattern is specified for engineering: user-facing name, SQL-style detection logic against your schema, an example user-facing insight string, the pre-trade gate recommendation (HARD BLOCK vs SOFT NUDGE), and frequency/severity estimate.

6.1 Revenge Spiral

Detection logic:

```
sql

WITH recent AS (
  SELECT user_id, trade_id, entry_time, win_loss, net_pnl,
         quantity * entry_price AS notional,
         LAG(win_loss, 1) OVER (PARTITION BY user_id ORDER BY entry_time) AS prev1,
         LAG(win_loss, 2) OVER (PARTITION BY user_id ORDER BY entry_time) AS prev2,
         LAG(exit_time, 1) OVER (PARTITION BY user_id ORDER BY entry_time) AS prev_exit_time,
         AVG(quantity * entry_price) OVER (PARTITION BY user_id
                                           ORDER BY entry_time ROWS BETWEEN 30 PRECEDING AND 1 PRECEDING) AS median_notional
  FROM trades)
SELECT * FROM recent
WHERE prev1='L' AND prev2='L'
      AND notional >= 1.5 * median_notional
      AND EXTRACT(EPOCH FROM entry_time - prev_exit_time)/60 < 30
      AND (emotion_entry IN ('revenge','anxious','fomo') OR trade_type='impulsive');
```

Insight shown: "You've taken 2 losses in the last 30 minutes and your size on this entry is 1.8× your normal. 73% of your trades that match this pattern have closed at -2R or worse." **Pre-trade gate:** **HARD BLOCK** — require 15-minute cooling-off OR explicit override with typed confirmation. This is the single highest-conviction block; it directly maps to FTMO's #1 failure cause. **Frequency / severity:** ~5-8% of trades for at-risk users; severity 4-5R per occurrence.

6.2 Stop-Loss Removal

Detection logic:

```
sql

SELECT trade_id FROM trades
WHERE stop_loss_defined = TRUE
      AND stop_loss_moved = 'widened'
      AND ABS(exit_price - entry_price)/entry_price >
          ABS(planned_stop_loss - entry_price)/entry_price * 1.3;
```

(Requires `planned_stop_loss` schema addition; without it, fall back to `stop_loss_moved='widened'` flag alone.) **Insight shown:** "You widened your stop on this trade and exited 1.8× past your original stop. Across your last 47 stop-widens, average loss is 2.4× planned risk." **Pre-trade gate: HARD BLOCK on the modification action** — when user attempts to widen stop, surface a one-click "accept the planned loss" button as friction. **Frequency / severity:** 10–20% of red trades; severity multiplier of 2–4× planned R.

6.3 Snap-Cut Winner (Disposition)

Detection logic:

```
sql

WITH user_stats AS (
  SELECT user_id,
    AVG(CASE WHEN win_loss='W' THEN hold_minutes END) AS avg_win_hold,
    AVG(CASE WHEN win_loss='L' THEN hold_minutes END) AS avg_loss_hold,
    AVG(CASE WHEN win_loss='W' THEN r_multiple END) AS avg_win_r
  FROM trades GROUP BY user_id HAVING COUNT(*) >= 30)
SELECT user_id FROM user_stats
WHERE avg_win_hold / NULLIF(avg_loss_hold,0) < 0.7
  OR avg_win_r < 0.6; -- planned R typically 1+
```

Insight shown: "Your average winner is held 38 min; your average loser, 142 min. Extending winners by 0.5R historically would have lifted your monthly expectancy from +0.04R to +0.21R."

Pre-trade gate: SOFT NUDGE at exit attempt when `r_multiple < planned_r * 0.6` and price still trending favorably: "You're exiting at 0.4R; your plan was 1.2R. Are you exiting on plan or on emotion?" with one-click "stick to plan." **Frequency / severity:** persistent in ~50% of users; cumulative drag 2–5% annualized.

6.4 Averaging Into Pain

Detection logic:

```
sql

SELECT child.trade_id FROM trades child
JOIN trades parent
  ON parent.user_id = child.user_id
  AND parent.instrument_name = child.instrument_name
  AND parent.direction = child.direction
  AND parent.entry_time < child.entry_time
  AND parent.exit_time IS NULL -- still open
WHERE (child.direction='long' AND child.entry_price < parent.entry_price)
  OR (child.direction='short' AND child.entry_price > parent.entry_price);
```

Insight shown: "You're adding to a position that's underwater. Of your last 22 averaged-down trades, 18 closed at -3R or worse. Adding to losers has cost you ₹X this year." **Pre-trade gate: HARD BLOCK** for `(asset_class IN ('fno', 'crypto', 'forex'))` with leverage > 5× — the blow-up mathematics are unforgiving. **(Acy) SOFT NUDGE** for unleveraged equity. **Frequency / severity:** 3-10% of position-opens for affected users; severity high (canonical blow-up signature).

6.5 Streak Tilt-Up

Detection logic:

```
sql

WITH s AS (
  SELECT *, LAG(win_loss,1) OVER w AS p1, LAG(win_loss,2) OVER w AS p2,
           LAG(win_loss,3) OVER w AS p3,
           quantity*entry_price AS notional,
           AVG(quantity*entry_price) OVER (
             PARTITION BY user_id ORDER BY entry_time
             ROWS BETWEEN 30 PRECEDING AND 1 PRECEDING) AS med_notional
  FROM trades WINDOW w AS (PARTITION BY user_id ORDER BY entry_time))
SELECT trade_id FROM s
WHERE p1='W' AND p2='W' AND p3='W'
      AND notional >= 1.3 * med_notional;
```

Insight shown: "You're 3-for-3 today and sizing up 1.6×. Historically, your 4th trade after a 3-win streak loses 64% of the time at this size. Consider keeping risk constant." **Pre-trade gate: SOFT NUDGE** with one-tap "keep size the same." **Frequency / severity:** ~3% of trades; severity 3-5R when it fires (gives back the streak).

6.6 Closing-Bell Make-Back / Cycle-End Gambling

Detection logic:

```
sql

SELECT trade_id FROM trades t
WHERE session = 'closing'
      AND EXTRACT(HOUR FROM entry_time) IN (last 60 min of session)
      AND (SELECT SUM(net_pnl) FROM trades t2
           WHERE t2.user_id=t.user_id AND t2.entry_date=t.entry_date
           AND t2.entry_time<t.entry_time) < 0
      AND quantity*entry_price >= 1.5 *
           AVG(quantity*entry_price) OVER (PARTITION BY user_id ROWS 30 PRECEDING);
-- Cycle-end variant: AND day_of_prop_cycle IN (last 3)
```

Insight shown: *"It's the last 45 minutes and you're red ₹X today. Your last-hour trades on red days are profitable 22% of the time."* **Pre-trade gate:** **HARD BLOCK** for prop-firm users in last 3 days of evaluation cycle when daily loss > 2%; **SOFT NUDGE** otherwise. **Frequency / severity:** ~15% of red days for active users; severity 3–5R, frequently the daily-loss-limit breach.

6.7 FOMO Chase

Detection logic:

```
sql

SELECT trade_id FROM trades
WHERE trade_type='impulsive'
      OR emotion_entry='fomo'
      OR (followed_plan='no' AND
          entry_price_pctile_in_prior_60min_bar > 0.85 AND direction='long')
      OR (followed_plan='no' AND
          entry_price_pctile_in_prior_60min_bar < 0.15 AND direction='short');
-- entry_price_pctile requires market-data join; fallback uses self-tag
```

Insight shown: *"You marked this trade impulsive and entered near the top of the 1-hour range. Your last 19 FOMO entries averaged -1.4R."* **Pre-trade gate:** **SOFT NUDGE** — checkbox at entry: "Is this on your plan?" If `followed_plan='no'`, surface 30-second pause + setup quality score. **Frequency / severity:** 8–15% of trades; cumulative drag dominates many accounts.

6.8 Boredom Trades

Detection logic:

```
sql

SELECT trade_id FROM trades
WHERE session='midday'
      AND market_condition='ranging'
      AND conviction_level <= 2
      AND emotion_entry='bored'
      AND hold_minutes < 0.5 *
          AVG(hold_minutes) OVER (PARTITION BY user_id);
```

Insight shown: *"3 of your 4 lunch-hour trades this week were tagged boredom or low-conviction. They're costing you 0.3R per trade in fees alone."* **Pre-trade gate:** **SOFT NUDGE** when `conviction_level<=2` AND `market_condition='ranging'`: *"Skip this and revisit in 30 min?"* **Frequency / severity:** persistent 5–10% of trades; severity low individually, high cumulatively (matches SEBI 27% cost-of-loss finding).

6.9 Disposition Inversion (account-level)

Detection logic: as 6.3 — but reported as a *standing diagnostic* on the dashboard rather than a

per-trade gate. **Insight shown:** "Your hold-time ratio (winners/losers) is 0.34 — well below the 0.7 threshold where disposition effect typically destroys edge. Estimated annual drag: 3.2% of capital." **Pre-trade gate:** None (account-level insight). **Frequency / severity:** affects ~50% of users; chronic drag.

6.10 Theta Gambler (market-gated: India F&O & US options)

Detection logic:

```
sql

SELECT trade_id FROM trades
WHERE instrument_type='option'
  AND (expiry_date - entry_date) <= 2
  AND (
    -- long premium late in day
    (option_type IN ('CE', 'PE', 'CALL', 'PUT') AND direction='long'
      AND EXTRACT(HOUR FROM entry_time) >= session_last_2h)
    OR
    -- naked short premium without defined-risk wrapper into event
    (direction='short' AND
      quantity*strike_price > 5 * ABS(net_pnl_at_entry) -- notional >> premium
      AND scheduled_event_within_24h = TRUE)
  );
```

Insight shown (long): "You're buying ODTE in the last 2 hours. Across retail, long ODTE loses ~\$8 per contract on average (Cboe 2023). Your own record: -₹X over 31 trades." **Insight shown (short):** "You're short premium with notional 8× your collected credit, holding into RBI policy. Single-day blow-up risk is the dominant failure mode here." **Pre-trade gate:** **HARD BLOCK** for naked short premium into scheduled events (RBI/Fed/earnings); **SOFT NUDGE** for long ODTE. **Frequency / severity:** ~40% of Indian F&O users on Thursdays; severity catastrophic on event days.

Step 7 — Redundant and overlapping patterns to merge

Several literature loops are different surface expressions of the same root cause and should be presented to users as **one pattern with multiple triggers** to avoid noise.

Merge target	Loops merged into it	Rationale
Revenge Spiral (6.1)	Loop 1 + Loop 10 (EOD make-back, when triggered after losses) + portion of Loop 13 (system abandonment when triggered by drawdown)	All driven by break-even effect; surface as one pattern with sub-types: <i>intraday revenge</i> , <i>closing-bell revenge</i> , <i>cycle-end revenge</i>
Hold-Time Asymmetry (6.3 + 6.9)	Loop 3 (premature winners) + Loop 6 (break-even anchor) + Loop 11 (account-level disposition)	All disposition-effect manifestations; one user-facing insight with two cuts (per-trade nudge + account diagnostic)
Conviction-Action Mismatch	Loop 5 (FOMO) + Loop 8 (hesitation-chase) + Loop 9 (boredom)	All off-plan entries with <code>followed_plan='no'</code> or <code>trade_type='impulsive'</code> ; show as one umbrella "Off-Playbook" diagnostic, with per-occurrence flavor (FOMO vs boredom vs late-chase) determined by <code>emotion_entry</code>
Sizing Volatility	Loop 4 (hot-streak) + Loop 1 (revenge size spike) + portion of Loop 14 (leverage trap)	All measure deviation of position size from user's own rolling median; one underlying "Sizing Discipline Score" surfaces three triggers

Net result: 10 distinct Tier-1 patterns collapse cleanly into **8 user-facing product surfaces** (Revenge Spiral, Stop Removal, Hold-Time Asymmetry, Averaging Into Pain, Sizing Discipline, Off-Playbook Entry, Closing-Bell/Cycle Risk, Theta Gambler) — well within the cognitive load budget.

Step 8 — Missing data requirements for full Tier-1 detection

Six of the ten Tier-1 patterns degrade gracefully without these additions; four require them for sharp detection. Listed in priority order.

- `planned_stop_loss` (price) and `planned_target` (price) at trade entry. Required for sharp Stop Removal (6.2), Snap-Cut Winner (6.3), and Hesitation-Chase. Without these, you fall back to user-tagged "what went wrong" labels — much noisier. *Add as required fields in pre-trade plan capture.*
- `planned_trigger_price` (the level at which the user intended to enter). Required for Hesitation-Chase detection (entry_price vs planned_trigger_price slippage). *Add to pre-trade plan capture.*

3. **account_equity_snapshot** at trade open. Required for normalizing position size as % of equity (rather than raw notional) — without this, every revenge / sizing detection is fragile when account size changes. *Pull from broker integration or daily user-input snapshot.*
4. **Broker-side order_modifications** log (timestamps and old/new stop prices). Required for in-flight stop-widening detection beyond user self-tagging. *Capture via broker API where available (Zerodha Kite, IBKR, MT4/cTrader logs, Binance/Bybit order amend events).*
5. **Market-context fields:** **atr_14**, **funding_rate_pctile**, **vol_index_at_entry**, **scheduled_event_within_24h** flag. Required for FOMO chase (entry-percentile in bar range), Leverage-Funding trap, and Theta Gambler event flag. *Pull from market data API; cache hourly.*
6. **prop_firm_account** flag + **cycle_start_date** + **daily_loss_limit_pct** + **max_drawdown_pct**. Required to surface FTMO-specific gates (cycle-end risk, DLL proximity warnings). *Add as account-level metadata when user links a prop firm account.*
7. **screenshot_at_entry** (optional). Useful for setup-quality grading post-hoc and BIKB ("but I knew better") trade detection. Defer to V2.
8. **pre_trade_plan_text** capture. A 30-second free-text "why I'm taking this trade" before order entry — major signal for off-playbook detection and post-trade review. Cheap to add.

Step 9 — Final consolidated output

9.1 Top 10 V1 patterns — final spec table

#	Pattern (user-facing)	Behavioral root cause	Schema fields used	Financial impact	Implementation logic
1	Revenge Spiral	Break-even effect; loss aversion	win_loss (lag), quantity , entry_price , entry_time , emotion_entry , trade_type , followed_plan	4-5R per occurrence; #1 FTMO DLL breach cause	Sequence detector + size- vs-rolling- median + emotion overrid
2	Stop Removal	Loss aversion; illusion of control	stop_loss_defined , stop_loss_moved , exit_price , <i>planned_stop_loss</i>	2-4× planned R loss; max-DD risk	Diff(exit, planned_stop) c widen events

#	Pattern (user-facing)	Behavioral root cause	Schema fields used	Financial impact	Implementatio logic
3	Hold-Time Asymmetry (umbrella)	Disposition effect	<code>hold_minutes</code> , <code>win_loss</code> , <code>r_multiple</code> , <i>planned_target</i>	2-5% annual drag (Odean) <code>(SciSpace +3)</code>	Account-level ratio + per-trade nudge at sub- target exits
4	Averaging Into Pain	Sunk cost; risk- seeking in losses	<code>instrument_name</code> , <code>direction</code> , <code>entry_time</code> , <code>entry_price</code> (sequence)	Canonical blow-up; Indian F&O / forex grid / crypto leverage	Same-instrume adds while pare leg negative
5	Sizing Discipline (umbrella: streak + revenge + leverage)	House money; <code>(Semantic Scholar)</code> overconfidence; break-even effect	<code>quantity</code> , <code>entry_price</code> , <code>leverage</code> , <code>win_loss</code> (lag), <i>account_equity</i>	Single trade gives back streak gains; trailing-DD risk	Position-size CV + delta after streaks/losses
6	Off-Playbook Entry (umbrella: FOMO + boredom + hesitation- chase)	Affect heuristic; fear-of-missing- out; boredom	<code>trade_type</code> , <code>emotion_entry</code> , <code>followed_plan</code> , <code>conviction_level</code> , <code>setup_type</code> , <i>entry- vs-planned-trigger</i>	8-15% of trades; 30- 50% expectancy haircut	Self-tagged + b percentile + conviction filter
7	Closing-Bell / Cycle-End Risk	Break-even effect; mental accounting; deadline pressure	<code>entry_time</code> , <code>session</code> , daily P&L sum, <i>prop_cycle_day</i>	DLL breach; #1 prop- firm fail-day	Last-hour large- size on red days final 3 cycle day w/ deficit
8	Theta Gambler	Lottery preference; probability weighting	<code>instrument_type</code> , <code>expiry_date</code> , <code>option_type</code> , <code>direction</code> , <code>strike_price</code> , <code>quantity</code>	Long ODTE: - \$8/contract; <code>(University of Lancaster)</code> <code>(Banana Farmer)</code> naked short premium: catastrophic gap	DTE filter + last 2h or pre-event check
9	Leverage- Funding	Overconfidence; herding	<code>leverage</code> , <code>asset_class</code> ,	98%+ liquidation at 125x; <code>(arXiv +2)</code> Oct-	Top-decile leverage + top-

#	Pattern (user-facing)	Behavioral root cause	Schema fields used	Financial impact	Implementatio logic
	Trap (crypto/forex)		<i>funding_rate_pctile</i>	2025 cascade \$20B retail (Solidus Labs)	decile funding/(
10	Regime Drift	Narrative fallacy; failure to update (AMH)	(strategy_id), (setup_type), (market_condition), (r_multiple), (win_loss) over rolling 30	10-20% drawdown on previously-winning system	Rolling SQN dro + concentration in one market_conditi

9.2 Patterns to REMOVE (low signal/noise — Tier 3)

- **Identity-Defense Hold** — requires social/public-commitment data we don't have; weak detection.
- **Trauma-Anniversary Freeze** — requires multi-year history and somatic state; cold-start hostile.
- **Confirmation Scroll** — requires browser/attention telemetry outside scope of a journaling app.
- **Hindsight-Distorted Review** — NLP on free-text journaling notes; expensive, low actionability.
- **Gambler's-Fallacy Mean-Reversion** — overlaps heavily with Off-Playbook Entry; not separable from data.

9.3 Patterns to MERGE

- **Premature exit + Break-even anchor + Account disposition** → single **Hold-Time Asymmetry** pattern (per-trade nudge + account diagnostic).
- **Hot-streak oversize + Revenge size spike + Leverage abuse** → single **Sizing Discipline** umbrella with three sub-triggers (streak, post-loss, leverage spike).
- **FOMO chase + Boredom trade + Hesitation-then-chase** → single **Off-Playbook Entry** umbrella with (emotion_entry) flavor differentiating the sub-type.
- **EOD make-back + Cycle-end gambling + System abandonment-post-drawdown** → single **Closing-Bell / Cycle-End Risk** with a multi-horizon framing.
- **News-event slippage + Leverage-funding trap** → single **Leverage-Funding Trap** (event-flag is one of the triggers).

Executive recommendation layer

V1 launch — minimum viable pattern set

Ship 8 user-facing patterns (#1-#8 in the table above). Drop Leverage-Funding Trap and Regime Drift to V1.5 (4-8 weeks post-launch) — both require either market-data integrations (funding rate, OI percentile) or a 30+ trade history per strategy that early users won't have. Eight patterns is the correct Goldilocks count: enough to be marketable as comprehensive, few enough that each insight retains weight and doesn't become wallpaper.

Suggested rollout order

The sequence is dictated by **detection sharpness** × **cold-start tolerance**, not by impact alone:

1. **Hold-Time Asymmetry** (account-level diagnostic) — *ships day 1*. Works on any history of 30 trades; immediate "this is me" moment. No schema changes needed.
2. **Off-Playbook Entry** — *ships day 1*. Self-tagged data (`emotion_entry`), (`followed_plan`), (`trade_type`) — no market data dependency.
3. **Sizing Discipline** — *ships day 1*. Pure structural data. Highest "wow" insight per user (sizing variance is shockingly visible to users who've never seen it charted).
4. **Revenge Spiral** — *week 2*. Needs ~30 trades for personalized median; HARD BLOCK requires careful UX.
5. **Closing-Bell / Cycle-End Risk** — *week 2*. Activate fully once prop-firm account linkage exists.
6. **Stop Removal** — *week 4*. Needs (`planned_stop_loss`) schema field rolled out and broker-API listening for amends.
7. **Averaging Into Pain** — *week 4*. Position-sequence detection requires reliable parent-leg matching.
8. **Theta Gambler** — *week 6*. India F&O and US options markets require expiry-calendar and event-calendar feeds.

Pre-trade gate philosophy

Default to soft nudges; reserve hard blocks for two specific conditions only. A global discipline product fails the moment users feel babysat — they will turn off notifications or churn. The gate philosophy:

- **Hard block only** when (a) the action will likely produce immediate prop-firm rule violation (DLL breach, max-DD breach), or (b) the action is the literal blow-up signature (averaging into a leveraged loser, naked short premium pre-event). Hard blocks should always be overridable via typed confirmation ("type 'I accept the risk' to proceed") — the friction is the intervention, not the prohibition. This preserves user autonomy and avoids feature-rejection.

- **Soft nudge** for everything else: 30-second pause + a single pre-computed sentence using the user's own historical data ("Your last 19 FOMO entries averaged -1.4R"). Personalized historical evidence outperforms generic warnings by an order of magnitude in behavior-change literature.
- **Account-level diagnostics** (Hold-Time Asymmetry, Regime Drift) are **passive insights**, not gates. They appear in weekly review.

The asymmetry is intentional: **2 hard blocks, 6 soft nudges, 2 passive diagnostics** across V1.

Schema additions strongly recommended BEFORE launch

Five additions unlock half the Tier-1 patterns; do not ship without them:

1. `planned_stop_loss` (price, decimal) — captured at entry time as required field.
2. `planned_target` (price, decimal) — captured at entry; powers Hold-Time Asymmetry sharpness.
3. `planned_trigger_price` (price, decimal) — captures the level at which user intended to enter (vs actual fill).
4. `account_equity_snapshot` (decimal, currency) — daily snapshot, normalizes everything else.
5. `pre_trade_plan_text` (varchar 500) — 30-second free-text "why I'm taking this trade." Powers off-playbook detection AND becomes the post-trade review anchor.

Two further additions (`prop_firm_account` flag with cycle metadata; market-context fields ATR/funding/event-flag) unlock V1.5 patterns and should follow within the first 60 days.

Cold-start delays — Tier-1 patterns to defer

Three patterns are technically Tier-1 but should be **soft-launched in shadow mode** (computed, logged, but not surfaced to users) until per-user data thickens:

1. **Account-level Disposition / Hold-Time Asymmetry** — needs ≥ 30 trades for stable PGR/PLR. Show a placeholder "build your history (12/30)" until threshold met.
2. **Revenge Spiral with personalized median** — uses rolling-30-trade median for size baseline. For first 30 trades, fall back to absolute % of equity threshold (e.g., position > 3% of equity after 2 losses).
3. **Regime Drift** — needs 30+ trades per `strategy_id` per `market_condition` to be statistically meaningful; otherwise produces false positives within normal variance. Defer entirely to V2 with explicit "needs 60 days of history" framing.

Marketing positioning

The eight V1 patterns directly address the documented destruction mechanisms in the empirical

record: **89–93% F&O loss rate (SEBI)**, **(Mostly Economics)** **74–89% CFD loss rate (ESMA)**, **(Complyport)** **67% forex loss rate (CFTC)**, **73–81% crypto retail loss rate (BIS)**, **(SuperCryptoNews)** **(RinggitPlus)** **and the FTMO/Topstep ~90% evaluation failure rate.** **(AquaFunded)** The marketing claim is defensible: *"Every pattern LatentEdge surfaces traces to peer-reviewed evidence of the specific mechanism by which retail traders lose money."* That is the line. It is true. It will sell.

What this product is NOT trying to do

LatentEdge is not a strategy generator, not a signal service, and not a coach replacement. It is a **discipline scoreboard with calibrated friction**. The patterns above all share one property: they are *behavioral, not predictive*. The product wins when users execute their own existing edge more cleanly — not when they discover a new edge. That positioning protects the product from the central failure mode of every "trading psychology" tool to date: scope creep into market commentary, where it competes with everything and wins nothing.